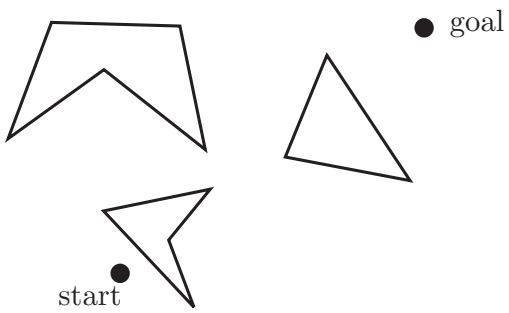
Homework

Robotics (B-KUL-H02A4A)

2025-2026

1. Draw a 3-DOF robot arm in a configuration where the inverse position kinematics has: a. a unique solution, b. exactly two solutions, c. infinitely many solutions. You can draw this on your computer/tablet or on paper and then take a (clear!) picture. For each case: explain why the number of solutions is as stated, and describe what geometric property causes this behavior. Briefly motivate your answer.

2. Describe an algorithm that drives the unicycle robot / rolling coin from any arbitrary initial configuration in its four-dimensional C-space to any arbitrary goal configuration, despite the two nonholonomic constraints. The control inputs are the rolling speed θ and the turning speed ϕ. Explain clearly in words or pseudocode how the algorithm would work. It is not necessary to give actual code or formulas.

3. Find the exponential coordinates $\hat{\omega}\theta \in \mathbb{R}^3$ for the $SO(3)$ matrix $\begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

4. Implement a complete path planner for the point robot with three polygonal obstacles. Choose your own values for the coordinates of all vertices, such that they approximately resemble the situation sketched in the attachment. Make your own implementation in Python. You can use basic libraries like `numpy`, but you cannot use specialized third-party libraries. Hint: check slides 24 and 25 of Lecture 5 for pseudocode and the overall flow. Include the full source code. You can use code you find on the internet / public repositories, but we expect that you can explain all code you include.